

Report conversation

Given a XML file with this format:

```
<robot name="iCub">
  <gazebo reference="r_lower_leg">
    <sensor name="r_lower_leg_mtb_acc_11b10"
type="imu">
      <always_on>1</always_on>
      <update_rate>100</update_rate>
      <pose>0.027894499999999982 -0.0422139
-0.043468000000000001 1.8325955062747417
-1.5707963267948963 0.0</pose>
      <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
    </sensor>
  </gazebo>
  <sensor name="r_lower_leg_mtb_acc_11b10"
type="accelerometer">
    <parent link="r_lower_leg"/>
    <origin rpy="1.8325955062747417
-1.5707963267948963 0.0" xyz="0.027894499999999982
-0.0422139 -0.043468000000000001"/>
  </sensor>
  <gazebo reference="r_lower_leg">
    <sensor name="r_lower_leg_mtb_acc_11b11"
type="imu">
      <always_on>1</always_on>
      <update_rate>100</update_rate>
      <pose>0.0164801 0.042248999999999997
-0.092958000000000004 -1.5707936954714847
-1.5706802103471817 -0.26180181049562956</pose>
      <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
    </sensor>
  </gazebo>
  <sensor name="r_lower_leg_mtb_acc_11b11"
type="accelerometer">
    <parent link="r_lower_leg"/>
```

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<origin rpy="-1.5707936954714847
-1.5706802103471817 -0.26180181049562956"
xyz="0.0164801 0.042248999999999997
-0.092958000000000004"/>
</sensor>
<gazebo reference="r_foot">
  <sensor name="r_foot_mtb_acc_11b12" type="imu">
    <always_on>1</always_on>
    <update_rate>100</update_rate>
    <pose>0.103479399999999997
-0.0008121999999999999 -0.0012149999999999966
1.4648752297108978e-16 2.2204452113283224e-16
-1.5707960430308967</pose>
    <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
  </sensor>
</gazebo>
<sensor name="r_foot_mtb_acc_11b12"
type="accelerometer">
  <parent link="r_foot"/>
  <origin rpy="1.4648752297108978e-16
2.2204452113283224e-16 -1.5707960430308967"
xyz="0.103479399999999997 -0.0008121999999999999
-0.0012149999999999966"/>
</sensor>
<gazebo reference="r_foot">
  <sensor name="r_foot_mtb_acc_11b13" type="imu">
    <always_on>1</always_on>
    <update_rate>100</update_rate>
    <pose>0.078479399999999999
-0.0118122000000000009 -0.0012149999999999966
2.220446049250314e-16 2.220446049250313e-16
-1.5707963267948966</pose>
    <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
  </sensor>
</gazebo>
<sensor name="r_foot_mtb_acc_11b13"
type="accelerometer">
  <parent link="r_foot"/>
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    <origin rpy="2.220446049250314e-16
2.220446049250313e-16 -1.5707963267948966"
xyz="0.07847939999999999 -0.011812200000000009
-0.001214999999999966"/>
  </sensor>
  <gazebo reference="l_upper_leg">
    <sensor name="l_upper_leg_ems_acc_eb6"
type="imu">
      <always_on>1</always_on>
      <update_rate>100</update_rate>
      <pose>-0.012577000000000012
-0.040407200000000002 -0.08340699999999998
1.5707963267948972 1.5707963267948961 0.0</pose>
      <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
    </sensor>
  </gazebo>
  <sensor name="l_upper_leg_ems_acc_eb6"
type="accelerometer">
    <parent link="l_upper_leg"/>
    <origin rpy="1.5707963267948972
1.5707963267948961 0.0" xyz="-0.012577000000000012
-0.040407200000000002 -0.08340699999999998"/>
  </sensor>
  <gazebo reference="l_upper_leg">
    <sensor name="l_upper_leg_ems_acc_eb10"
type="imu">
      <always_on>1</always_on>
      <update_rate>100</update_rate>
      <pose>-0.012575000000000003
0.040406799999999998 -0.040226999999999985
-1.5707963267948968 -1.5707963267948963 0.0</pose>
      <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
    </sensor>
  </gazebo>
  <sensor name="l_upper_leg_ems_acc_eb10"
type="accelerometer">
    <parent link="l_upper_leg"/>
    <origin rpy="-1.5707963267948968

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-1.5707963267948963 0.0"
xyz="-0.012575000000000003 0.040406799999999998
-0.0402269999999999985"/>
</sensor>
<gazebo reference="l_upper_leg">
  <sensor name="l_upper_leg_ems_gyro_eb6"
type="imu">
  <always_on>1</always_on>
  <update_rate>100</update_rate>
  <pose>-0.013081000000000016
-0.040407200000000002 -0.087979 1.5707963267948972
1.5707963267948961 0.0</pose>
</sensor>
</gazebo>
<sensor name="l_upper_leg_ems_gyro_eb6"
type="gyroscope">
  <parent link="l_upper_leg"/>
  <origin rpy="1.5707963267948972
1.5707963267948961 0.0" xyz="-0.013081000000000016
-0.040407200000000002 -0.087979"/>
</sensor>
<gazebo reference="l_upper_leg">
  <sensor name="l_upper_leg_ems_gyro_eb10"
type="imu">
  <always_on>1</always_on>
  <update_rate>100</update_rate>
  <pose>-0.013079 0.040406799999999998
-0.0356549999999999964 -1.5707963267948968
-1.5707963267948963 0.0</pose>
</sensor>
</gazebo>
<sensor name="l_upper_leg_ems_gyro_eb10"
type="gyroscope">
  <parent link="l_upper_leg"/>
  <origin rpy="-1.5707963267948968
-1.5707963267948963 0.0" xyz="-0.013079
0.040406799999999998 -0.0356549999999999964"/>
</sensor>
<gazebo reference="l_upper_leg">
  <sensor name="l_upper_leg_mtb_acc_10b1"

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type="imu">
  <always_on>1</always_on>
  <update_rate>100</update_rate>
  <pose>0.042001689999999994
0.031564799999999999 0.049131999999999998
-1.5707955185655114 -1.5706802103355728
-0.7330390947129777</pose>
  <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
</sensor>
</gazebo>
<sensor name="l_upper_leg_mtb_acc_10b1"
type="accelerometer">
  <parent link="l_upper_leg"/>
  <origin rpy="-1.5707955185655114
-1.5706802103355728 -0.7330390947129777"
xyz="0.042001689999999994 0.031564799999999999
0.049131999999999998"/>
</sensor>
<gazebo reference="l_upper_leg">
  <sensor name="l_upper_leg_mtb_acc_10b2"
type="imu">
  <always_on>1</always_on>
  <update_rate>100</update_rate>
  <pose>0.053521149999999998
0.0077657000000000014 0.039252000000000001
-1.5707963863768455 -1.5706802103927162
-1.3089969473954748</pose>
  <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
</sensor>
</gazebo>
<sensor name="l_upper_leg_mtb_acc_10b2"
type="accelerometer">
  <parent link="l_upper_leg"/>
  <origin rpy="-1.5707963863768455
-1.5706802103927162 -1.3089969473954748"
xyz="0.053521149999999998 0.0077657000000000014
0.039252000000000001"/>
</sensor>

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<gazebo reference="l_upper_leg">
  <sensor name="l_upper_leg_mtb_acc_10b3"
type="imu">
  <always_on>1</always_on>
  <update_rate>100</update_rate>
  <pose>0.053784999999999986
0.0150706000000000003 0.00418199999999999635
3.141592653589793 -1.5707963267948961 0.0</pose>
  <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
  </sensor>
</gazebo>
<sensor name="l_upper_leg_mtb_acc_10b3"
type="accelerometer">
  <parent link="l_upper_leg"/>
  <origin rpy="3.141592653589793
-1.5707963267948961 0.0" xyz="0.053784999999999986
0.0150706000000000003 0.00418199999999999635"/>
  </sensor>
<gazebo reference="l_upper_leg">
  <sensor name="l_upper_leg_mtb_acc_10b4"
type="imu">
  <always_on>1</always_on>
  <update_rate>100</update_rate>
  <pose>0.05378499999999998
-0.0144294000000000002 0.00418199999999999635
3.141592653589793 -1.5707963267948961 0.0</pose>
  <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
  </sensor>
</gazebo>
<sensor name="l_upper_leg_mtb_acc_10b4"
type="accelerometer">
  <parent link="l_upper_leg"/>
  <origin rpy="3.141592653589793
-1.5707963267948961 0.0" xyz="0.05378499999999998
-0.0144294000000000002 0.00418199999999999635"/>
  </sensor>
<gazebo reference="l_upper_leg">
  <sensor name="l_upper_leg_mtb_acc_10b5"

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type="imu">
  <always_on>1</always_on>
  <update_rate>100</update_rate>
  <pose>0.034129549999999998 -0.0414999
-0.0048580000000000029 1.9547688910448342
-1.5707963267948961 0.0</pose>
  <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
</sensor>
</gazebo>
<sensor name="l_upper_leg_mtb_acc_10b5"
type="accelerometer">
  <parent link="l_upper_leg"/>
  <origin rpy="1.9547688910448342
-1.5707963267948961 0.0" xyz="0.034129549999999998
-0.0414999 -0.0048580000000000029"/>
</sensor>
<gazebo reference="l_upper_leg">
  <sensor name="l_upper_leg_mtb_acc_10b6"
type="imu">
  <always_on>1</always_on>
  <update_rate>100</update_rate>
  <pose>-0.043785 0.017810999999999994
-0.026410999999999999 -1.5707963267919405
1.5707743139371833 1.570796326789853</pose>
  <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
  </sensor>
</gazebo>
<sensor name="l_upper_leg_mtb_acc_10b6"
type="accelerometer">
  <parent link="l_upper_leg"/>
  <origin rpy="-1.5707963267919405
1.5707743139371833 1.570796326789853"
xyz="-0.043785 0.017810999999999994
-0.026410999999999999"/>
</sensor>
<gazebo reference="l_upper_leg">
  <sensor name="l_upper_leg_mtb_acc_10b7"
type="imu">

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    <always_on>1</always_on>
    <update_rate>100</update_rate>
    <pose>-0.04378500000000001
-0.014191800000000011 -0.02641099999999999
1.5707963267982468 1.5706473388946958
-1.5707963267919158</pose>
    <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
  </sensor>
</gazebo>
<sensor name="l_upper_leg_mtb_acc_10b7"
type="accelerometer">
  <parent link="l_upper_leg"/>
  <origin rpy="1.5707963267982468
1.5706473388946958 -1.5707963267919158"
xyz="-0.04378500000000001 -0.014191800000000011
-0.02641099999999999"/>
</sensor>
<gazebo reference="l_lower_leg">
  <sensor name="l_lower_leg_ems_acc_eb7"
type="imu">
    <always_on>1</always_on>
    <update_rate>100</update_rate>
    <pose>0.04351379199999999
0.0113144999999999963 -0.09276499999999999
-3.1415926535897922 -1.3526297834539558
-8.476849710455595e-16</pose>
    <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
  </sensor>
</gazebo>
<sensor name="l_lower_leg_ems_acc_eb7"
type="accelerometer">
  <parent link="l_lower_leg"/>
  <origin rpy="-3.1415926535897922
-1.3526297834539558 -8.476849710455595e-16"
xyz="0.04351379199999999 0.0113144999999999963
-0.09276499999999999"/>
</sensor>
<gazebo reference="l_lower_leg">

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    <sensor name="l_lower_leg_ems_gyro_eb7"
type="imu">
    <always_on>1</always_on>
    <update_rate>100</update_rate>
    <pose>0.04450335399999999
0.0118184999999999968 -0.08830099999999996
-3.1415926535897922 -1.3526297834539558
-8.476849710455595e-16</pose>
    </sensor>
</gazebo>
    <sensor name="l_lower_leg_ems_gyro_eb7"
type="gyroscope">
    <parent link="l_lower_leg"/>
    <origin rpy="-3.1415926535897922
-1.3526297834539558 -8.476849710455595e-16"
xyz="0.04450335399999999 0.0118184999999999968
-0.088300999999999996"/>
    </sensor>
    <gazebo reference="l_lower_leg">
    <sensor name="l_lower_leg_mtb_acc_10b8"
type="imu">
    <always_on>1</always_on>
    <update_rate>100</update_rate>
    <pose>0.053133529999999984
0.017184099999999994 -0.0043179999999999885
-2.8972464875193467 -1.5707963267948963 0.0</pose>
    <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
    </sensor>
</gazebo>
    <sensor name="l_lower_leg_mtb_acc_10b8"
type="accelerometer">
    <parent link="l_lower_leg"/>
    <origin rpy="-2.8972464875193467
-1.5707963267948963 0.0" xyz="0.053133529999999984
0.017184099999999994 -0.0043179999999999885"/>
    </sensor>
    <gazebo reference="l_lower_leg">
    <sensor name="l_lower_leg_mtb_acc_10b9"
type="imu">

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    <always_on>1</always_on>
    <update_rate>100</update_rate>
    <pose>0.054451899999999984
-0.0137991000000000009 -0.0043179999999999985
-1.5707963664504991 -1.5706802104435666
-1.919861927351539</pose>
    <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
  </sensor>
</gazebo>
  <sensor name="l_lower_leg_mtb_acc_10b9"
type="accelerometer">
  <parent link="l_lower_leg"/>
  <origin rpy="-1.5707963664504991
-1.5706802104435666 -1.919861927351539"
xyz="0.054451899999999984 -0.0137991000000000009
-0.0043179999999999985"/>
  </sensor>
  <gazebo reference="l_lower_leg">
    <sensor name="l_lower_leg_mtb_acc_10b10"
type="imu">
      <always_on>1</always_on>
      <update_rate>100</update_rate>
      <pose>0.025915199999999996
0.042744799999999997 -0.043468000000000001
-1.5707936954714847 -1.5706802103471817
-0.26180181049562956</pose>
      <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
    </sensor>
  </gazebo>
  <sensor name="l_lower_leg_mtb_acc_10b10"
type="accelerometer">
  <parent link="l_lower_leg"/>
  <origin rpy="-1.5707936954714847
-1.5706802103471817 -0.26180181049562956"
xyz="0.025915199999999996 0.042744799999999997
-0.043468000000000001"/>
  </sensor>
  <gazebo reference="l_lower_leg">

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    <sensor name="l_lower_leg_mtb_acc_10b11"
type="imu">
    <always_on>1</always_on>
    <update_rate>100</update_rate>
    <pose>0.018459299999999977
-0.0417187000000000025 -0.092958000000000004
1.5707953192782331 -1.5707699978643246
0.26180018702487956</pose>
    <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
    </sensor>
</gazebo>
    <sensor name="l_lower_leg_mtb_acc_10b11"
type="accelerometer">
    <parent link="l_lower_leg"/>
    <origin rpy="1.5707953192782331
-1.5707699978643246 0.26180018702487956"
xyz="0.018459299999999977 -0.0417187000000000025
-0.092958000000000004"/>
    </sensor>
    <gazebo reference="l_foot">
    <sensor name="l_foot_mtb_acc_10b12" type="imu">
    <always_on>1</always_on>
    <update_rate>100</update_rate>
    <pose>0.10552739999999997
0.00081220000000000129 -0.001214999999999966 0.0
-0.0 1.5707963267948966</pose>
    <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
    </sensor>
</gazebo>
    <sensor name="l_foot_mtb_acc_10b12"
type="accelerometer">
    <parent link="l_foot"/>
    <origin rpy="0.0 -0.0 1.5707963267948966"
xyz="0.10552739999999997 0.00081220000000000129
-0.001214999999999966"/>
    </sensor>
    <gazebo reference="l_foot">
    <sensor name="l_foot_mtb_acc_10b13" type="imu">

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    <always_on>1</always_on>
    <update_rate>100</update_rate>
    <pose>0.08052739999999997
0.011812200000000002 -0.0012149999999999966 0.0 -0.0
1.5707963267948966</pose>
    <plugin name="iCub_yarp_gazebo_plugin_ACC"
filename="libgazebo_yarp_inertial.so"/>
  </sensor>
</gazebo>
<sensor name="l_foot_mtb_acc_10b13"
type="accelerometer">
  <parent link="l_foot"/>
  <origin rpy="0.0 -0.0 1.5707963267948966"
xyz="0.08052739999999997 0.011812200000000002
-0.0012149999999999966"/>
</sensor>
<gazebo reference="r_hip_2">
  <sensor name="r_upper_leg_ft_acc_3b11" type="imu">
    <always_on>1</always_on>
    <update_rate>100</update_rate>
    <pose>0.006563800000000002
-0.0085861000000000013 -0.059281999999999946
-2.220446049250314e-16 2.220446049250313e-16
1.5707963267948966</pose>
  </sensor>
</gazebo>
<sensor name="r_upper_leg_ft_acc_3b11"
type="accelerometer">
  <parent link="r_hip_2"/>
  <origin rpy="-2.220446049250314e-16
2.220446049250313e-16 1.5707963267948966"
xyz="0.006563800000000002 -0.0085861000000000013
-0.059281999999999946"/>
</sensor>
<gazebo reference="r_hip_2">
  <sensor name="r_upper_leg_ft_gyro_3b11"
type="imu">
    <always_on>1</always_on>
    <update_rate>100</update_rate>
    <pose>0.006563800000000002

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-0.0085861000000000013 -0.0592819999999999946
-2.220446049250314e-16 2.220446049250313e-16
1.5707963267948966</pose>
  </sensor>
</gazebo>
  <sensor name="r_upper_leg_ft_gyro_3b11"
type="gyroscope">
  <parent link="r_hip_2"/>
  <origin rpy="-2.220446049250314e-16
2.220446049250313e-16 1.5707963267948966"
xyz="0.0065638000000000002 -0.0085861000000000013
-0.0592819999999999946"/>
</sensor>
<gazebo reference="r_ankle_2">
  <sensor name="r_foot_ft_acc_3b14" type="imu">
    <always_on>1</always_on>
    <update_rate>100</update_rate>
    <pose>0.0040000999999999926
-0.008600000000000001 -0.0452820000000000044
-2.220446049250314e-16 2.220446049250313e-16
1.5707963267948966</pose>
  </sensor>
</gazebo>
  <sensor name="r_foot_ft_acc_3b14"
type="accelerometer">
  <parent link="r_ankle_2"/>
  <origin rpy="-2.220446049250314e-16
2.220446049250313e-16 1.5707963267948966"
xyz="0.0040000999999999926 -0.008600000000000001
-0.0452820000000000044"/>
</sensor>
<gazebo reference="r_ankle_2">
  <sensor name="r_foot_ft_gyro_3b14" type="imu">
    <always_on>1</always_on>
    <update_rate>100</update_rate>
    <pose>0.0040000999999999926
-0.008600000000000001 -0.0452820000000000044
-2.220446049250314e-16 2.220446049250313e-16
1.5707963267948966</pose>
  </sensor>

```

```

</gazebo>
<gazebo>
  <pose>0.0 0.0 0.63 0.0 0.0 3.14</pose>
</gazebo>
</robot>

```

Can you write a Python script with lxml that remove all the sensor tags that have name
r_upper_leg_ft_gyro_3b11 or l_foot_mtb_acc_10b12 ?
Thanks!

Certainly! You can use the `lxml` library in Python to parse and manipulate the XML file. Here's a sample script that removes the specified sensor tags:

python

 Copy code

```

from lxml import etree

# Load the XML file
tree = etree.parse('your_xml_file.xml')
root = tree.getroot()

# Define the sensor names to be removed
sensor_names_to_remove = ['r_upper_leg_ft_gyro_3b11', 'l_foot_mtb_acc_10b12']

# Find and remove the specified sensor tags
for sensor_name in sensor_names_to_remove:
    for sensor_tag in root.xpath(f"//sensor[@name='{sensor_name}']"):
        sensor_tag.getparent().remove(sensor_tag)

# Save the modified XML back to a file
tree.write('modified_xml_file.xml', pretty_print=True)

```